Decode ways 18

A message containing digits from '0' to '9' is encoded into letters from 'A' to 'Z' using the mapping 'A' → "1", 'B' → "2", ..., 'Z' → "26".

You are given a string of digits. You need to find the number of valid ways to decode it back into letters. However, there is a special rule for a decoding to be considered valid: the decoded string must not contain any two adjacent vowels. For this problem, the vowels are 'A', 'E', 'I', 'O', and 'U'.

Leading zeros make a number grouping invalid (e.g., "06" cannot be decoded to 'F'). Since the answer may be large, return the number of valid decodings modulo $10^9 + 7$.

Input Format

The first line contains a string, s, denoting the [message. The string] consists only of digits from '0' to '9'.

Constraints

1 <= |s| <= 10^5

Output Format

The output contains an integer value

Sample Input 0

```
12
```

Sample Output 0

```
2
```

Explanation 0

Valid decodings are "AB" (1 -> A, 2 -> B) and "L" (12 -> L). The sequence "AB" has a vowel 'A' and a consonant 'B', so no adjacent vowels. The sequence "L" has only a consonant. Both are valid. The answer is 2.

Sample Input 1

```
11
```

Sample Output 1

```
1
```

Explanation 1

The possible decodings are "AA" (1 -> A, 1 -> A) and "K" (11 -> K). The sequence "AA" has two adjacent vowels, so it is invalid. The sequence "K" has no adjacent vowels, making it valid. The answer is 1.

Sample Input 2

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151
```

Sample Output 2

```
0
```

Explanation 2

The possible decodings are "AEA" (1 -> A, 5 -> E, 1 -> A) and "OA" (15 -> O, 1 -> A). The combination "1" and "51" is invalid because 51 is out of range. Both "AEA" and "OA" contain adjacent vowels, so they are invalid. The answer is 0.